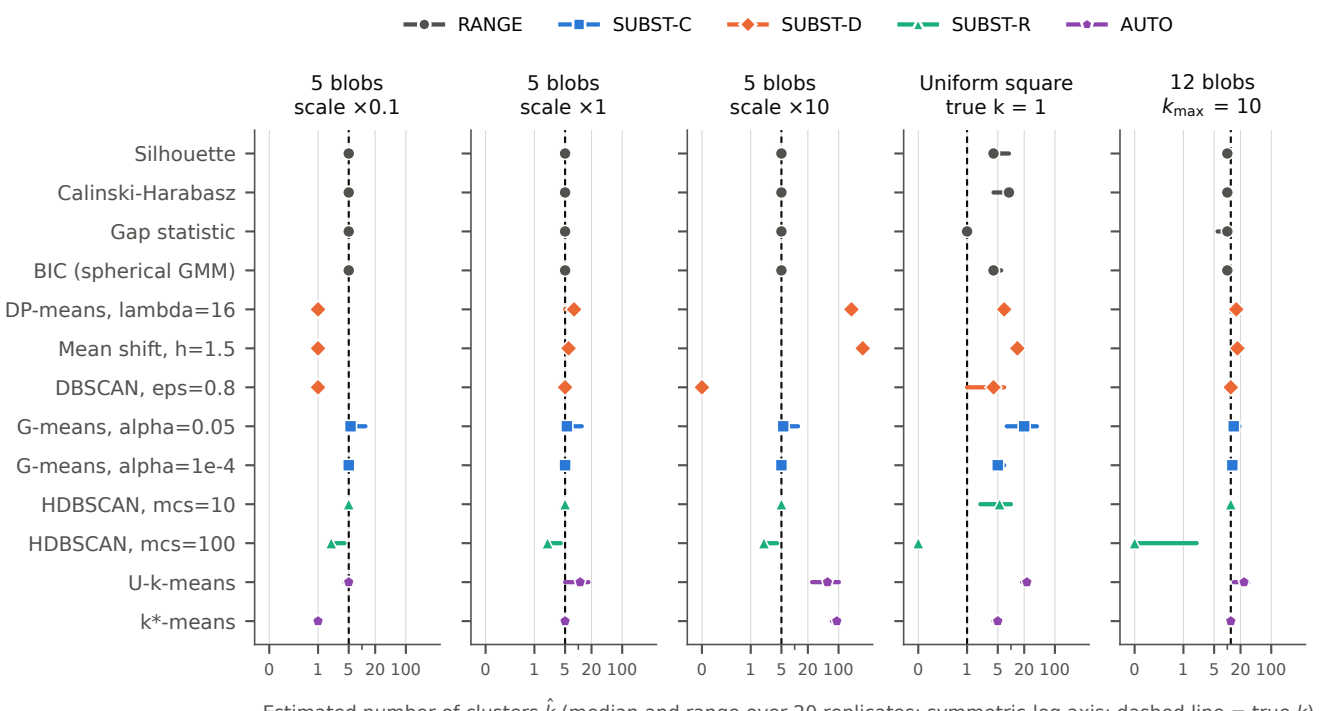
RANGE
SUBST-C
SUBST-D
SUBST-R
AUTO
5 blobs scale ×0.1
5 blobs scale ×1
5 blobs scale ×10
Uniform square true k = 1
12 blobs k_{max} = 10
Silhouette
Calinski-Harabasz
Gap statistic
BIC (spherical GMM)
DP-means, lambda=16
Mean shift, h=1.5
DBSCAN, eps=0.8
G-means, alpha=0.05
G-means, alpha=1e-4
HDBSCAN, mcs=10
HDBSCAN, mcs=100
U-k-means
k*-means
0 1 5 20 100
Estimated number of clusters $\hat{k}$ (median and range over 20 replicates; symmetric-log axis; dashed line = true k)